

Problem Statement Document

Project Title: IBM BOB Problem Statement #22: AI Interview Trainer Agent

Domain: Artificial Intelligence / Career Guidance & Skill Assessment

Platform: IBM watsonx Orchestrate & IBM Cloud

1. Background & Context

In today's competitive job market, candidates often struggle to prepare effectively for technical, behavioral, and HR interviews. Traditional interview preparation relies heavily on static question lists or expensive mock interview coaching. Candidates lack personalized, interactive practice sessions that provide real-time, objective feedback tailored to their specific experience level, domain, and target roles.

2. Problem Statement

Job seekers require an intelligent, interactive, and personalized preparation tool that can simulate real-world interview scenarios across multiple domains (Technical, Behavioral, HR, and Behavioral/Experience-level) while offering actionable, structured feedback to improve candidate performance.

Existing solutions fall short because they:

- Lack adaptation based on candidate skill level, domain, and target job roles.
- Provide generic or static questions without domain-specific context.
- Do not evaluate responses with detailed feedback, missing key concepts, or standardized scoring metrics.

3. Proposed Solution: AI Interview Trainer Agent

The **AI Interview Trainer Agent** is built on **IBM watsonx Orchestrate** and **IBM Cloud** to deliver an interactive, context-aware mock interview experience. The agent dynamically generates role-appropriate questions, retrieves knowledge-based reference answers, and evaluates user responses using a standardized scoring system.

Key Capabilities & Features:

- **Personalized Onboarding:** Tailors question domain, difficulty, and context using candidate profile data (role, experience level, key skills) via `candidate_profile.py`.
- **Knowledge Base (RAG) Integration:** Integrates domain-specific interview preparation materials using a Retrieval-Augmented Generation (RAG) pipeline (`knowledge_base/rag_knowledge.py`) to provide accurate, real-world context.

- **Structured Evaluation & Scoring:** Uses `prompt_templates.py` and `interview_engine.py` to evaluate candidate answers on a 1–10 scale using clear Markdown evaluation tables, highlighting strengths, gaps, and improvement tips.
- **Interactive Session Flow:** Conducts interviews one question at a time, supporting interactive commands such as skip and quit to simulate natural conversation flow.

4. Technical Architecture & Component Breakdown

- **Platform:** IBM watsonx Orchestrate
- **LLM Engine:** Watsonx Orchestrate Frontier Model
- **Core Business Logic (Python 3.12):**
 - `candidate_profile.py` — Manages candidate profile data and session state.
 - `interview_engine.py` — Handles question generation, answer evaluation, and score calculation.
 - `prompt_templates.py` — Defines evaluation rubrics, domain guides, and agent prompts.
 - `watsonx_client.py` — Manages direct API interaction with watsonx services.
 - `workspace_config.yaml` — Configures environment settings and agent parameters.

5. Expected Impact & Outcomes

- Provides equal access to high-quality, domain-specific interview training for students and professionals.
- Delivers objective, immediate, and actionable feedback to accelerate candidate learning cycles.
- Demonstrates seamless integration of IBM watsonx Orchestrate agent capabilities with custom Python tooling and RAG knowledge bases.